**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 243 · Year 13 autumn mock: Paper 1

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Evaluate true AND false.

Q2. Why should a model omit irrelevant details?

Q3. What does the Program Counter hold?

Q4. A full adder receives A=1, B=1 and carry-in=1. State sum and carry-out and justify using binary addition.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

diagnostic — Assessment used to find knowledge gaps.

two's complement — Encoding signed binary integers.

De Morgan's law — Negate operands and swap AND/OR.

register — Fast CPU storage for a specific role.

## New learning

Purpose — This is an original diagnostic Paper 1 practice, not an official OCR examination paper. Some systems topics will be retaught later.

Manage time — Allow approximately 15 minutes to each 15-mark question.

Show methods — Write the inversion/add-one steps, state every Boolean simplification rule and use precise CPU register terminology.

## Worked example

Unmarked example: what earns method marks for a negative 8-bit value?

1. Predict the method — Write +5 in exactly 8 bits.

2. Check and explain — Invert each bit, then add one to obtain the two's complement result for −5.

# We Do · Practise with guidance

Unmarked different example: evaluate ¬(A ∧ B) for A=1, B=0.

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A ∧ B = 1 AND 0 = 0
¬0 = 1
```

Predict the method

First calculate A AND B = 0.

Check and explain

Negate it: ¬0=1. Include intermediate reasoning when a question asks for it.

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**2. Write a complete new answer in GREEN PEN. Show working if needed:**





BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

(a) Convert −18 to eight-bit two's complement, showing working. [4] (b) Perform arithmetic right shift by 1 on that representation, giving the binary result and signed decimal value. [3] (c) Add $1011_2$ and $0110_2$, showing carry and answer. [3] (d) Normalise positive $0.0011_2 \times 2^5$, showing shifts and decimal value. [5]

### Question 2

(a) Complete four output rows of A XOR B in order 00,01,10,11. [4] (b) Simplify ¬(A ∧ B) using De Morgan's law. [3] (c) For a half adder with A=1, B=1, state Sum and Carry and explain the gates. [4] (d) Simplify A ∨ (B ∧ ¬B) using Boolean identities. [4]

### Question 3

(a) State the roles of PC, MAR, MDR, CIR and ACC. [5] (b) Describe a simplified fetch-decode-execute cycle using PC, MAR, MDR, CIR and the control unit. [6] (c) Explain one role of an interrupt and an interrupt service routine. [4]

## Exit Ticket · Answer before checking

For an 8-bit signed representation, adding a ninth bit automatically gives the correct 8-bit answer.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.



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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. False.

Q2. To reduce complexity while retaining data needed to answer the question.

Q3. The address of the next instruction to fetch.

Q4. 1+1+1=3 decimal=11 binary; Sum=1, carry-out=1.

### You Do pages · Independent questions

Q1. MARK SCHEME /15. (a) +18=00010010 [1], invert 11101101 [1], add 1=11101110 [2]. (b) Sign-preserving right shift gives 11110111 [2] = −9 [1]. (c) Correct binary addition yields $10001_2$ [2], correct carry evidence [1]. (d) $0.0011\times2^5 \rightarrow 0.0110\times2^4$ [2] $\rightarrow 0.1100\times2^3$ [2], decimal 6 [1].

Q2. MARK SCHEME /15. (a) Outputs 0,1,1,0 [4]. (b) ¬A ∨ ¬B [2], De Morgan swaps AND to OR and negates components [1]. (c) Sum=0 [1], Carry=1 [1], XOR gives sum [1], AND gives carry [1]. (d) B∧¬B=0 [2]; A∨0=A [2].

Q3. MARK SCHEME /15. (a) PC address of next instruction [1], MAR memory address accessed [1], MDR data/instruction transferred [1], CIR current instruction [1], ACC result/intermediate arithmetic [1]. (b) PC address into MAR [1]; instruction fetched from memory into MDR [1]; PC advanced [1]; instruction copied to CIR [1]; control unit decodes [1]; processor executes and updates registers as needed [1]. (c) Interrupt signals event requiring attention [1]; CPU checks/services it at a suitable point [1]; relevant ISR handles event [1]; execution state restored and normal processing resumes [1].

### Exit Ticket · Model answer

False. Preserve the specified width and account for overflow.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**